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An Initial examination of fear of negative and positive evaluation in youth[☆]

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ABSTRACT

Fear of negative evaluation (FNE) and fear of positive evaluation (FPE) are both core features of social anxiety. The majority of research with these constructs has been done with older adolescents and adults, with only one previous study examining FNE and FPE in childhood. However, this previous work relied exclusively on parent-report of youth FNE and FPE. Here, we examined the factor structure of FNE and FPE using youth self-reports. Moreover, we examined the associations with dimensions of internalizing and externalizing problems. We found that two-factor structure of FNE and FPE was a marginal fit to the data. Exploratory models identified three items that showed significant cross-loadings on non-target factors. Overall, we found that FNE was associated with dimensions of internalizing problems reported by youth and their mothers. FPE was associated with internalizing problems reported by youth, but not parents. Associations between FNE and clinical outcomes were stronger than those for FPE. This study demonstrates promise of FNE and FPE in youth and highlights important directions for future research.

1. Introduction

Fear of negative evaluation (FNE) is a core construct within social anxiety and social anxiety disorder (e.g., Clark & Wells, 1995). More recently, there has been an accumulation of evidence showing that fear of positive evaluation (FPE) is also integral in the understanding of social anxiety (e.g., Weeks, Heimberg, & Rodebaugh, 2008). The roles of FNE and FPE have been extensively examined in older adolescents and adults (e.g., Reichenberger, Wiggert, Wilhelm, Weeks, & Blechert, 2015; Rodebaugh, Weeks, Gordon, Langer, & Heimberg, 2012). There is a smaller, but established literature on FNE in children; yet, the literature on FPE in children is only beginning (e.g., Poole, Hassan, & Schmidt, 2022). Towards building this literature on FPE in children, we examine the internal structure of FNE and FPE and associations between FNE and FPE and important clinical correlates using child self-reports.

FNE has long been considered a central construct in social anxiety. Fears of possible negative consequences from others based on anticipated negative feedback challenges individuals to engage or be successful in multiple occupational, peer, and family situations. FPE shares the anticipated negative outcomes of occupational, peer, and family situations, but does so when individuals anticipate receiving positive feedback from others. The anticipation and possibility of receiving positive attention may lead to increased conspicuousness (i.e., feeling "in the spotlight"), anxious affect, and avoidance of situations. FNE has been assessed using a small number of measures (Carleton, Collimore, & Asmundson, 2007; Rodebaugh et al., 2004; Watson & Friend, 1969; Weeks et al., 2005; Weeks, Heimberg, Rodebaugh et al., 2008). The Fear of Negative Evaluation (FNE) scale was the earliest developed (Watson & Friend, 1969), which was followed by the Brief Fear of Negative Evaluation (BFNE) scale (Carleton et al., 2007; Leary, 1983; Rodebaugh et al., 2004; Weeks et al., 2005). FPE has been assessed using only a single instrument, the Fear of Positive Evaluation Scale (FPES; Weeks et al., 2008; but see also Wilson, Cassin, & Antony, in press). In psychometric analyses, the use of reverse scored items led to diminished reliability and understanding of the items on these assessments among participants (see Rodebaugh et al., 2004; Weeks et al., 2005). Thus, much of the work conducted using these measures has relied solely on the forward-scored items.

A central question about the utility of the FPE construct is whether it provides incremental explanation of social anxiety symptoms. Cook,

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Moore, Bryant, and Phillips (2022) conducted a meta-analysis of the associations between FNE and FPE and social anxiety symptoms. They found that inclusion of FPE explained an additional 9% of variance in social anxiety symptoms relative to FNE in isolation. Thus, for understanding the processes involved in the maintenance of social anxiety, both FNE and FPE warrant attention.

Despite sharing overlap in fear of evaluation, factor analytic studies identify these as correlated, but distinct constructs. Weeks, Norton, and Heimberg (2009) found that FNE and FPE were represented as unidimensional constructs – in this study, the latent correlation between the FNE and FPE factors was.74. In a meta-analysis, Cook et al. (2022) reported the association between FNE and FPE was.48. Thus, across studies and using their sum score composites, the constructs were evidenced to be moderately correlated.

The work on FNE and FPE has extensively examined adults and older adolescents. Studies examining these constructs in youth samples are only beginning to be reported. Social fears typically emerge during childhood and change in intensity and structure throughout development (e.g., Bokhorst, Westenberg, Oosterlaan, & Heyne, 2008). Given the developmental changes that occur during childhood and adolescence (e.g., social cognition and increased importance of peers; Andrews, Ahmed, & Blakemore, 2021; Kilford, Garrett, & Blakemore, 2016; Laursen & Veenstra, 2021), FNE and FPE are important to examine during this period, both to understand more about the development of fear of evaluation and to support prevention of the negative consequences associated with these constructs (e.g., greater anxiety and avoidance of socialization). In the initial study, Poole et al. (2022) administered the BFNE and FPES to parents of youth age 8–9 years (to obtain *parent* ratings of the extent to which their children tend to experience FNE and FPE). They examined the internal structure of the measure and found that the correlated two-factor structure was a superior fit to a single factor model. The association between FNE and FPE was.49. These results are consistent with findings of studies of older adolescents and adults. The authors also examined associations between parent-reported FNE and FPE and parent-reported social anxiety. They found that FNE and FPE were both independently associated with social anxiety. Although this is an important and promising start to examining the utility of both FNE and FPE in youth samples, the reliance on parent-report of FNE and FPE is an important limitation to this work. Moreover, there is speculation that FNE develops before FPE (Fredrick & Luebbe, 2020). Thus, the associations between FNE and FPE with important clinical outcomes may not be parallel in childhood but become more similar later in adolescence and adulthood.

As FNE originated directly from work on social anxiety, there are strong associations between these constructs. There are also reported associations between FNE and related internalizing problem constructs, such as broader anxiety symptoms (Shafique, Gul, & Raseed, 2017) and depression (O'Connor, Berry, Weiss, & Gilbert, 2002). Results for FPE, however, are more mixed. There are associations between FPE and social anxiety, with those tending to be more modest, but explaining additional variance beyond FNE (Wang, Hsu, Chiu, & Liang, 2012; Weeks, 2015). Examination of associations between FNE or FPE and externalizing problem dimensions are rarer in the literature. However, the development of the nomological networks for FNE and FPE have been developed in samples of adults or adult informants of youth FNE and FPE.

There have been no previous studies of youth *self*-responses to measures assessing FNE and FPE. This study began addressing this line of work. We examines the structure of self-reported FNE and FPE using confirmatory factor analysis (CFA) with the full set of items, as well as the items that included only the straight-forwardly worded items (i.e., items which would be expected to directly index the construct in question) and not the reverse-scored items (i.e., items which were designed with the intent of indexing the absence or opposite of the construct in question, for the purpose of disrupting/detecting response biases [e.g., see Weeks, Weeks, Heimberg, Rodebaugh, & Norton et al., 2005; 2008]). We also examines associations between identified dimensions of FNE and FPE and youth self- and parent-reports of youth anxiety, depression, and externalizing problems. We include examination of associations with externalizing problems as tests of discriminant validity.

2. Methods

2.1. Participants

Participants were drawn from the first wave of the Temple Adolescent Development Study, a prospective three-year longitudinal study examining associations between development and reward function. This sample included English-speaking families from the Philadelphia community with a child between 9 and 14 years at baseline who resided with at least one biological parent. Exclusion criteria included child or parental history of bipolar disorder or psychotic spectrum disorder. Youth were also excluded if they were currently taking psychotropic medications (with the exception of medication for attention deficit hyperactivity disorder) or were diagnosed with a serious neurological illness, developmental delay, pervasive developmental disorder, autism spectrum disorder, dyslexia, or dysgraphia. Furthermore, in preparation for future study assessments, participants were screened at baseline for fMRI eligibility and were excluded if they presented with any conditions that would make participation in an fMRI unsafe (e.g., metal braces, retainers, pacemakers, vision difficulties, claustrophobia). Overall intellectual ability for children was also examined at the baseline assessment using the Kaufman Brief Intelligence Test (KBIT-2; (Kaufman, 1990), and participants with estimates of overall intelligence levels falling two or more SDs below the mean were excluded (KBIT-2 FSIQ < 70).

The full recruited sample included 249 adolescents (57%; 142 female) ages 9–14 ($M_{age} = 11.01$, $SD = 1.50$). After consenting, 17 families were excluded. For those 17 excluded at baseline, six were excluded due to family history of bipolar spectrum or psychotic disorders, one youth had no biological parent as an involved caretaker, one was discovered to be out of the age range, seven had cognitive function less than 70 on the KBIT-2, one youth had a documented Autism Spectrum Disorder diagnosis, and one youth had a tic disorder and history of concussion (which deemed them ineligible for other study assessments [e.g., MRI]). Due to the length of the assessment, self-report questionnaires were provided to youth and their parents to complete at home. However, participants were inconsistent in returning the packets (child return rate = 57.3–63.7%; parent return rate = 62.1–75.4%). Of these participants, 144 children (59% female) completed all self-report measures included in the analyses. As such, these participants and their parent(s) were included in the current study. Participants for the current study sample ranged from 8.48 to 14.37 years of age (*Mean [M]* = 11.05 years, *Standard Deviation [SD]* = 1.48). Fifty-three (36.8%) participants reported their race as Black/African American, 53 (36.8%) as White/Caucasian, 10 (6.9%) as Asian, 5 (3.5%) as multiracial, 3 (2.1%) as "Other," and 20 individuals did not report their race.

2.2. Procedure

For this study, we relied on the baseline questionnaire assessments completed by youth and their primary caregivers. Baseline questionnaires were provided to participants and their parents at the conclusion of the in-person assessments. There were no significant differences between participants who did and did not return questionnaire assessments on demographic characteristics. Given that the present study utilized measures that were originally designed to assess fear of evaluation in adults, and examined the psychometric properties of these revised measures in children (aged from 8.48 to 14.37 years of age; see above for details), we examined the readability of the assessments of FNE and FPE, and revised them to be appropriate for children (as well as adolescents). We calculated reading grade levels using the Flesch-

Kincaid Grade Level Formula (Kincaid et al., 1975).

2.3. Measures

2.3.1. Fear of negative evaluation

The original Brief Fear of Negative Evaluation Scale (BFNE; Leary, 1983) is a 12-item self-report questionnaire that assesses fear and distress related to negative evaluation from others. The original BFNE items are evidenced to have a reading level of 6.6 – thus, we edited the BFNE items to be more appropriate for children to permit the administration of the measure as a self-report. We hereby refer to our revised BFNE items as the BFNE-Children scale (BFNE-C; see Table 1 for revised items). Importantly, our revised BFNE-C items are evidenced to have a reading level of 4.4. There are eight straight-forwardly worded items; evidence from psychometric evaluations of the original BFNE support greater validity of the sum of the straight-forwardly worded items only (Rodebaugh et al., 2004; Weeks et al., 2005). As such, the current study utilized the eight straightforwardly-worded BFNE-C items to obtain an index of FNE, and omitted the reverse-scored items. Responses are rated on a 5-point scale from 1 (*not at all characteristic of me)* to 5 (*extremely characteristic of me*). In previous studies, the original BFNE has demonstrated strong convergent and discriminant validity (Collins, Westra, Dozois, & Stewart, 2005) and internal consistency (all αs > .92) in both undergraduate (Rodebaugh et al., 2004) and clinical (Weeks et al., 2005) adult samples. Initial research has also supported the use of the original BFNE in youth (i.e., adolescent) samples (Karp et al., 2018; pp, 2156; Lipton, Augenstein, Weeks, & De Los Reyes, 2014; Vagos et al., 2016). The BFNE-C demonstrated excellent internal consistency in the current child sample as well (α = .94).

2.3.2. Fear of positive evaluation

The original Fear of Positive Evaluation Scale (FPES; Weeks et al., 2008; Weeks, Heimberg, Rodebaugh, Goldin, & Gross, 2012) is a 10-item self-report questionnaire that assesses fear and distress related to positive evaluation from others. The original FPES items are evidenced to have a reading level of 8.5 – thus, we edited the FPES items to be more appropriate for children, and also removed an item which was selected for removal in a modification of the FPES for adolescents (see Lipton, Weeks, Daruwala, & De Los Reyes, 2016 for details). This was also done to permit youth providing self-reports. We hereby refer to our revised FPES items as the FPES-Children scale (FPES-C; see Table 1 for revised items) – importantly, our revised FPES items are evidenced to have a reading level of 3.9. Two reverse-scored items are included, but are not used in calculating the total score (see Weeks et al., 2008 for details). As such, the current study utilized seven items to calculate an index of FPE. Responses for the child version of the FPES are rated on a 5-point scale from 1 (*not at all characteristic of me)* to 5 (*extremely characteristic of me*). In previous studies, the FPES has demonstrated strong convergent and discriminant validity (Fergus et al., 2009; Weeks et al., 2008; Weeks, Heimberg, Rodebaugh et al., 2008) and internal consistency (all αs > .80) in both undergraduate (Weeks et al., 2008; Weeks et al., 2010) and clinical (Fergus et al., 2009; Weeks et al., 2012) adult samples. Initial research has also supported the use of the FPES in youth (i.e., adolescent) samples in the US (Karp et al., 2018; pp, 2156; Lipton et al., 2014) and Portugal (Vagos et al., 2016). The FPES-C demonstrated adequate internal consistency in the current sample as well (α = .74).

Table 1
BFNE-Child and FPES-Child Items.

	Item Prompt
BFNE-C1*	*I worry about what other people will think of me even when I know it doesn't make any difference.*
BFNE-C2	*I don't worry if people are forming a bad idea about me.*
BFNE-C3*	*I am always afraid that others will find out my flaws.*
BFNE-C4	*I don't worry that people will think badly of me.*
BFNE-C5*	*I'm afraid that others won't approve of me.*
BFNE-C6*	*I'm afraid that people will find fault with me.*
BFNE-C7	*When people think badly of me, it doesn't bother me.*
BFNE-C8*	*When I talk to someone, I worry they will think badly of me.*
BFNE-C9*	*I am always worried that people think badly of me.*
BFNE-C10	*If I know someone is judging me negatively, it has little effect on me.*
BFNE-C11*	*I tend to be nervous that people will think badly of me.*
BFNE-C12*	*I often worry that I will say or do the wrong things.*
FPES-C1*	*I don't like showing off in front of my peers.*
FPES-C2*	*I wouldn't want to be complimented by someone I like.*
FPES-C3*	*I don't like getting praise from grown-ups when I'm with my peers.*
FPES-C4	*I say things in class that I think are interesting.*
FPES-C5*	*If I get a compliment, I'd rather get it when I'm alone.*
FPES-C6*	*I wonder if I do things "too well" when my peers are watching.*
FPES-C7*	*I don't like it when people compliment me.*
FPES-C8*	*I don't like to be noticed, even when I am being admired.*
FPES-C9	*I wish people would comment more on my skills.*

* Straightforwardly-worded items; BFNE-C = Brief Fear of Negative Evaluation scale-Child version; FPES-Child = Fear of Positive Evaluation Scale-Child version

2.3.3. Anxiety symptoms

Children and parents reported on youth anxiety using the Screen for Child Anxiety Related Emotional Disorders (SCARED; Birmaher, 1977). The SCARED is a 41-item questionnaire that screens for several anxiety disorders, including panic disorder/somatic symptoms, generalized anxiety disorder, separation anxiety disorder, social anxiety disorder, and school avoidance. The SCARED total score screens for the presence of any anxiety disorder and was used as a measure of anxiety symptoms in the current study. Responses are rated on a 3-point scale from 0 (*not true or hardly true*) to 2 (*very true or often true*). In previous studies, the SCARED has demonstrated strong convergent and discriminant validity and internal consistency (αs $\geq$.90) in clinical (Birmaher et al., 1999) and non-clinical (Hale, Raaijmakers, Muris, & Meeus, 2005) samples. The SCARED demonstrated good internal consistency for child reports (α ranged from.61 [School Avoidance] to.87 [GAD]; ω ranged from.68 [School Avoidance] to.88 [GAD]) and parents (α ranged from.58 [School Avoidance] to.88 [Social Anxiety]; ω ranged from.62 [School Avoidance] to.83 [GAD]).

2.3.4. Depressive symptoms

Children and parents reported on youth depressive symptoms using the Children's Depression Inventory (CDI; Kovacs, 1992). The CDI is a 27-item self-report measure of children's depressive symptoms for the past two weeks. Responses each have three choices (e.g., "I am sad once in a while," "I am sad many times," or "I am sad all the time"), and participants are asked to choose the answer that best describes themselves. In previous studies, the CDI has demonstrated adequate convergent and discriminant validity and internal consistency (αs $\geq$.80) in clinical and non-clinical (Lee, Krishnan, & Park, 2012; Saylor, Finch, & Spirito, 1984) samples. The CDI demonstrated good internal consistency in the current sample as well in both children (α = .89, ω = .90) and parents (α = .84, ω = .86).

2.3.5. Irritability

As an index of youth self-report externalizing behaviors, youth provided reports of irritability using the Affective Reactivity Index (ARI; Stringaris, Zavos, Leibenluft, Maughan, & Eley, 2012). The ARI assesses youth's irritable mood over the past 6 months using six items rated on a 3-point scale from (never true) to (certainly true). The total score had strong internal consistency (α = .88, ω = .89).

2.3.6. Broad internalizing and externalizing problems

Mothers provided assessments of broad youth internalizing and externalizing problem using the Child Behavior Checklist. The Child Behavior Checklist (CBCL 6–18; Achenbach, 2001) is a parent-report measure that examines broad psychopathology in children over the last six months. The CBCL measure contains 119 items that are rated on a scale of 0 (*not true*) to 2 (*very true or often true*). The internalizing behavior subscale includes 32 items spanning anxious/depressed, withdrawn/depressed, and somatic complaints content. The externalizing behavior subscale includes 35 items spanning rule-breaking behavior and aggressive behavior content. The CBCL has demonstrated good reliability and validity (Achenbach & Edelbrock, 1979; Youngstrom, Loeber, & Stouthamer-Loeber, 2000) and is a valid screening instrument for externalizing (Hudziak, Copeland, Stanger, & Wadsworth, 2004) and internalizing (Ebesutani et al., 2010; Seligman, Ollendick, Langley, & Baldacci, 2004) disorders. Internal consistency estimates for parent report of youth internalizing and externalizing behaviors were excellent (ordinal alpha =.95 for internalizing and.97 for externalizing; ω = .89 for internalizing and.91 for externalizing).

2.4. Data analysis

All modeling was completed using Mplus 8.8 (Muthén & Muthén, 1998) and facilitated using the MplusAutomation (Hallquist & Wiley, 2018) package in R (R Core Team, 2022). Initial models were fit within a fully confirmatory factor analysis (CFA) framework, with a priori FNE items loading only on a single FNE factor, and with a priori FPE items loading only on a separate FPE factor. We also fit CFA models with only forward-scored items, as suggested by previous work with the FNE and FPE items (e.g., see Weeks et al., 2008). Given that this is an initial investigation of the use of the FNE and FPE items with youth providing self-reports, we also were open to conducting exploratory factor analyses in order to better understand the constructs.

All models were estimated using the robust weighted least squares estimator (WLSMV; Flora & Curran, 2004). This was done due to the ordinal response options for the FNE and FPE items. We evaluated models on three goodness of fit indices: the comparative fit index (CFI; Bentler, 1990), Root Mean Square Error of Approximation (RMSEA; Steiger, 1990), and Standardized Root Mean Squared Residual (SRMR). Although cut-offs are somewhat arbitrary (Marsh, Hau, & Wen, 2004), current conventions suggest that excellent model fit is indicated by CFI values $\geq .95$ (Hu & Bentler, 1999), RMSEA values $\leq .05$ (MacCallum, Browne, & Sugawara, 2006), and SRMR $< .06$; adequate fit is indicated by CFI $> .90$, RMSEA between.05 and.08, and SRMR $< .08$.

3. Results

3.1. Structure of FNE and FPE item sets

An initial CFA was estimated on the full itemset with FNE and FPE items loading on separate, correlated factors. All a priori FNE items loaded significantly on the specified factor. All but one a priori FPE items loaded significantly on the specified factor. The FNE and FPE factors were strongly correlated ($r = .64$, $p < .001$). However, this model was a poor fit to the data (Table 2).

Next, we fit the same two factor structure, but included only the straightforwardly worded items (i.e., removing the reverse scored items). All a priori FNE and FPE items loaded significantly on their specified factors. The FNE and FPE factors were strongly correlated ($r = .65$, $p < .001$). The CFI was excellent, however, the RMSEA and SRMR were not adequate (Table 2). Factor loadings are presented in Table 3.

Given the equivocal fit of our a priori hypothesized two-factor model, we also conducted exploratory factor analyses (EFA): [i] with all FPES-C and BFNE-C items (both straightforwardly worded and reverse scored) and [ii] with only the straightforwardly worded items from both measures. EFAs were specified using the oblique geomin rotation with 25 random starts. We estimated EFAs with up to six factors. Full model results are provided in the supplementary materials. For models that included all items (both straightforwardly worded and reverse scored) and that included only the straightforwardly worded items from both measures, adequate model fit based on CFI and RMSEA was found for solutions with two or more factors. Parallel analyses of the full item set (i.e., including both forward and reverse scored items) suggested that three factors should be retained, whereas for the straightforward items, only two factors should be retained. The interpretability of the factor solutions with three or more factors was unclear for both item sets considered. In the EFA model that included only straightforwardly worded items, model fit was good. Factor loadings are presented in Table 3. In this model, all eight a priori FNE items loaded significantly on the same factor and all seven a priori FPE items positively loaded significantly on the other factor. Two FPE items ("If I get a compliment, I'd rather get it when I'm alone" and "I wonder if I do things 'too well' when my peers are watching") had significant loadings on the FNE factor and one FNE item ("I worry about what other people will think of me even when I know it doesn't make any difference") also had a significant negative loading on the FPE factor.

3.2. Associations between FNE and FPE and Psychopathology Criteria

We estimated associations between the youth self-rated FNE and FPE factors and youth self- and parent-report of youth internalizing and externalizing problems. We estimated these associations by fixing the parameters for the measurement model of the two-factor EFA model and included correlations with the psychopathology correlates within a full exploratory structural equation model framework.

In these models, FNE was significantly associated with all measures of child self- and parent-report of youth internalizing and externalizing symptoms, except for youth self-report of irritability (Tables 4 and 5). For associations between FNE and youth self-reported psychopathology

Table 2
Model fit information.

Model	χ^2	df	CFI	RMSEA (90% CI)	SRMR
Two Factor CFA All Items	943.50	188	.808	.166 (.156-.177)	.185
Two Factor CFA Straightforwardly-Worded Items	218.84	89	.964	.100 (.084-.117)	.103
Two Factor EFA All Items	507.28	169	.915	.117 (.105-.129)	.127
Two Factor EFA Straightforwardly-Worded Items	141.67	76	.982	.077 (.057-.096)	.064

Table 3
Standardized actor loadings for the CFA and EFA of forwardly stated BFNE and BFPE items.

	CFA		EFA	
	FNE	FPE	FNE	FPE
BFNE-C1	0.674 ***		0.814 ***	-0.258 **
BFNE-C3	0.856 ***		0.856 ***	0.009
BFNE-C5	0.906 ***		0.968 ***	-0.128
BFNE-C6	0.928 ***		0.987 ***	-0.131
BFNE-C8	0.929 ***		0.842 ***	0.177
BFNE-C9	0.895 ***		0.872 ***	0.059
BFNE-C11	0.916 ***		0.808 ***	0.215
BFNE-C12	0.860 ***		0.774 ***	0.172
FPES-C1		0.415 ***	0.191	0.268 *
FPES-C2		0.602 ***	-0.081	0.755 ***
FPES-C3		0.800 ***	0.170	0.729 ***
FPES-C5		0.702 ***	0.250 *	0.530 ***
FPES-C6		0.885 **	0.513 ***	0.386 ***
FPES-C7		0.789 ***	0.068	0.846 ***
FPES-C8		0.603 ***	-0.062	0.738 ***

* $p < .05$;
* * $p < .01$;
* ** $p < .001$.

Table 4
ESEM correlations between youth reported FNE & FPE and child and parent reports of youth psychopathology.

	FNE	FPE	r FNE > r FPE
Child Self-Report			
SCARED Total	0.527 ***	0.240 **	**
Panic/Somatic	0.448 ***	0.137	**
GAD	0.506 ***	0.177 *	**
Separation Anxiety	0.373 ***	0.233 **	
Social Anxiety	0.379 ***	0.197 *	
School Avoidance	0.423 ***	0.203 *	*
CDI Total	0.461 ***	0.081	***
ARI Total	0.080	0.008	
Parent-Report			
SCARED Total	0.252 **	0.079	*
Panic/Somatic	0.158 *	-0.024	*
GAD	0.283 ***	0.007	**
Separation Anxiety	0.163 *	0.106	
Social Anxiety	0.192 *	0.128	
School Avoidance	0.156 *	0.079	
CDI Total	0.297 ***	0.041	**
CBCL Internalizing	0.363 ***	0.112	**
CBCL Externalizing	0.253 **	0.127	

* $p < .05$; * * $p < .01$; * ** $p < .001$. SCARED = Screen for Child Anxiety Related Disorders; GAD = Generalized Anxiety Disorder; CDI = Child Depression Inventory; ARI = Affective Reactivity Index; CBCL = Child Behavior Checklist.

Table 5
Simultaneous regression coefficients of FNE and FPE regressed on outcomes.

	FNE	FPE
Child Self-Report		
CDI Total	3.660 (0.699) ***	-0.800 (0.708)
ARI Total	0.414 (0.507)	-0.121 (0.523)
SCARED Total	6.898 (1.285) ***	0.629 (1.384)
Panic/Somatic	1.858 (0.380) ***	-0.152 (0.391)
GAD	1.889 (0.382) ***	-0.066 (0.372)
Separation Anxiety	1.074 (0.310) **	0.347 (0.316)
Social Anxiety	1.304 (0.385) **	0.229 (0.413)
School Avoidance	0.591 (0.120) ***	0.072 (0.142)
Parent-Report		
CDI Total	1.695 (0.469) ***	-0.433 (0.485)
SCARED Total	2.533 (0.863) ***	-0.186 (0.855)
Panic/Somatic	0.354 (0.140) *	-0.177 (0.188)
GAD	0.943 (0.293) **	-0.337 (0.277)
Separation Anxiety	0.418 (0.255)	0.151 (0.255)
Social Anxiety	0.581 (0.328)	0.222 (0.331)
School Avoidance	0.172 (0.090)	0.027 (0.095)
CBCL Internalizing	2.394 (0.469)***	-0.195 (0.564)
CBCL Externalizing	1.792 (0.678)**	0.272 (0.699)

* $p < .05$; * * $p < .01$; * ** $p < .001$. SCARED = Screen for Child Anxiety Related Disorders; GAD = Generalized Anxiety Disorder; CDI = Child Depression Inventory; ARI = Affective Reactivity Index; CBCL = Child Behavior Checklist.

dimensions, associations were of moderate magnitudes (*r*s ranged from.37–.53; without youth irritability considered). For associations between FNE and parent-reported youth psychopathology dimensions, associations were of more modest magnitudes (*r*s ranged from.16–.36).

In contrast, there were fewer significant associations between FPE and youth self- and parent-reported psychopathology dimensions. FPE was significantly associated with the SCARED Total score, along with the GAD, separation anxiety, social anxiety, and school avoidance subscales. FPE was not significantly associated with any dimensions of psychopathology reported by parents.

We tested for differences in the magnitude of correlations between dimensions of psychopathology and FNE and FPE. In these comparisons, we found that associations between FNE and child depression; SCARED total score, panic/somatic symptoms, and GAD for both youth self- and parent-reports; school avoidance for youth self-report; and internalizing problems for parent-report were stronger than those associations with FPE.

Finally, within the ESEM framework, we regressed each of the dimensions of psychopathology on the FNE and FPE factors as a simultaneous regression. In these models, the FNE factor was associated with all dimensions of psychopathology except for youth self-report of irritability and parent-report of youth social anxiety, separation anxiety, and school avoidance. There were no significant associations between FPE and any dimensions of psychopathology.

4. Discussion

The present study advances the literature on the bivalent fear of evaluation model of social anxiety by examining FPE and FNE among children aged 9–14 years. Indeed, the present study is the first that we are aware of that examined *self*-report responses pertaining to FPE in children. And similar to the only prior study examining FPE in children that has been published to date (Poole et al., 2022), a number of *parent*-report questionnaires were also examined. A number of noteworthy findings were obtained.

First, both the FPES-C and BFNE-C were structured to have a 4th grade reading level, and this was confirmed via the Flesch-Kincaid Grade Level Formula (Kincaid et al., 1975). Thus, both measures are evidenced to be appropriate for administration to children ages 8 years and up. However, as noted below, there may be some items and response options on the FNE and FPE scales that have ambiguous meanings for youth. Thus, there may be a need to revise items to improve their clarity in future work.

Second, confirmatory factor findings largely provided preliminary cross-validating evidence for the bivalent fear of evaluation model in a child sample. Namely, the a priori hypothesized bivalent fear of evaluation model that has been supported in adult (e.g., see Weeks et al., 2008) and adolescent (see Vagos et al., 2016) samples, wherein all straightforward FPES items load onto a single (FPE) factor and all straightforward BFNE items load onto a separate (FNE) correlated factor, showed minimal discrepancy between the data and the hypothesized model (CFI =.964) – indeed, model fit was excellent according to this specific index. There was, however, suboptimal prediction by the two-factor fear of evaluation model of the values that were observed in terms of error rates (RMSEA =.084–.117; SRMR =.103). However, to achieve good fit across all model fit indices, exploratory factor analyses revealed that this suboptimal prediction was likely due to the need to include cross-loadings for 3 out of the 15 items in the model (two FPES-C items, and one BFNE-C item), as the latter 3 items cross-loaded significantly on the alternative valence of fear of evaluation (i.e., FNE and FPE, respectively). Importantly, the factor loadings were strongest for our a priori hypothesized factors. Ergo, taken together in totality, this is suggestive that our factor analytic findings provide overall additional support for the bivalent fear of evaluation model of social anxiety within a sample of children. However, given the alteration of the a priori model, further work is needed to better understand the functioning of those items that loaded on both the FNE and FPE factors.

Third, children's self-reported FPE (i.e., FPES-C) and FNE (i.e., BFNE-C) responses were adequately internally consistent, similar to previous work in both adult (e.g., see Weeks et al., 2005; Weeks et al., 2008) and adolescent (e.g., see Lipton et al., 2014; Vagos et al., 2016) samples. In combination with the factor analytic results reviewed above, these results show that our measures of FPE and FNE for children (i.e., FPES-C and BFNE-C, respectively) are assessing specific constructs, reflected by responses to items on these scales. The more modest internal consistency estimates may be suggestive that the construct may crystallize with more maturation.

Fourth, similar to findings in adult and adolescent samples, FPE and FNE showed strongly and positively latent correlations ($r = .65$), further suggesting that FPE and FNE are hand-in-hand cognitive tendencies. Fifth, children's *self*-reported FPE and FNE symptoms related positively and significantly to social anxiety, separation anxiety, and school avoidance (see Table 3). In combination with the strong latent correlation between FPE and FNE (see above), this combination of findings provides some additional evidence that FPE and FNE are hand-in-hand cognitive components of social anxiety (e.g., see Weeks & Howell, 2012, 2014).

Sixth, consistent with previous findings in adults (e.g., see Wang et al., 2012; Weeks, 2015)(e.g., see Wang et al., 2012; Weeks, 2015), FPE was evidenced to be a fairly *unique* factor of social anxiety. To illustrate, although child self-reported FPE related significantly and positively to child self-reported social anxiety, separation anxiety (which is strongly related to social anxiety in children; e.g., see Essau, rer. soc, Conradt, Petermann, & Phil, 2000), and school refusal (which is strongly related to social anxiety in children; see Richards & Hadwin, 2011), child self-reported FPE did not relate significantly to self-reported panic/somatic symptoms, generalized anxiety symptoms, depressive symptoms, or irritability (see Table 3) – by contrast, further consistent with previous findings in adults (e.g., see Wang et al., 2012; Weeks, 2015), FNE was evidenced to be a *specific* (but not unique) factor of social anxiety in children in the present sample, in that FNE generally related more strongly to social anxiety than to other (non-social) anxiety symptoms and depressive symptoms, but nevertheless related significantly and positively to other (non-social) anxiety symptoms and depressive symptoms (see Wang et al., 2012). Indeed, child self-rated FNE related statistically-significantly and positively to nearly all other child- and parent-rated variables in the present study (with the exception of child self-rated irritability). Child self-rated FNE was more strongly associated with self-rated panic/somatic symptoms and generalized anxiety symptoms than with self-rated social anxiety symptoms; and, self-rated FNE related numerically more strongly to parent-rated generalized anxiety symptoms than to parent-rated social anxiety symptoms. Again, this is fairly consistent with research that has shown that FNE is a transdiagnostic anxiety factor (Reiss, 1997) that is essentially characteristic of all anxiety disorders and related conditions (e.g., depression).

Moreover, child self-rated FPE was not significantly related to any parent-rated symptoms. We had only minimal support for the relationship between child *self*-rated FPE and *parent*-rated social anxiety, as it was the strongest magnitude (albeit non-statistically-significant) association with child self-rated FPE. Furthermore, it bears noting that, upon simultaneously regressing FPE and FNE on outcomes, child self-rated FNE ceased to relate significantly to parent-rated social anxiety, separation anxiety (which is strongly related to social anxiety in children; e.g., see Essau et al., 2000), or school avoidance (which is strongly related to social anxiety in children; see Richards & Hadwin, 2011). This is consistent with previous research suggesting that FPE and FNE are hand-in-hand cognitive components of social anxiety (e.g., Weeks, 2022; Weeks & Howell, 2012, 2014). This is suggestive that shared variance in FNE and FPE may diminish unique associations that may be seen in separation and social anxiety, as well as school avoidance. Additionally, child-rated FPE demonstrated discriminant validity, in that FPE did not relate statistically significantly to parent-rated externalizing problems (although FNE did) or to child-rated irritability.

However, unexpectedly, child self-reported FPE did not account for unique variance in social anxiety symptoms (either self- or parent-rated) above and beyond FNE. This finding stands in stark contrast to previous findings in adult (e.g., see Weeks et al., 2008; Weeks et al., 2008; Weeks & Howell, 2012) and adolescent (see Karp et al., 2018, pp. 2156; Vagos et al., 2016) samples, wherein FPE has been reliably shown to account for unique variance in social anxiety symptoms above FNE. There are a number of explanations to this unexpected pattern of findings.

The most parsimonious explanation of FPE not accounting for unique variance in social anxiety beyond FNE within the present sample is that FPE is less characteristic of social anxiety in children. However, this latter explanation would be inconsistent with our findings that: (i) children's self-rated FPE related positively and significantly to self-rated social anxiety (as well as separation anxiety and school avoidance), and (ii) that children's self-rated FNE did not relate significantly more strongly to self-rated social anxiety in comparison to self-rated FPE (see Table 3).

An alternative explanation for FPE not accounting for unique variance in social anxiety beyond FNE within the present sample is that FPE may be experienced less saliently during childhood than during adolescence (e.g., see Karp et al., 2018, pp. 2156; Vagos et al., 2016) or adulthood (e.g., see Weeks et al., 2008). This latter possible explanation is consistent with the conclusions of Poole et al. (2022), who claimed that FNE and FPE are more closely related and share larger overlap in adolescence and adulthood compared to childhood. Perhaps children, at an earlier developmental stage, have had fewer opportunities to experience negative repercussions of positive social attention (Gilbert, 2014; Weeks, Menatti, & Howell, 2015)(e.g., social reprisal from perceived more powerful and dominant peers; e.g., see Gilbert, 2014; Weeks et al., 2015), and thus, FPE is relatively less prominent for socially anxious children than for socially anxious adults and adolescents. There are also developmental changes into adolescence where the importance and salience of peers increases (Andrews et al., 2021; Kilford et al., 2016; Laursen & Veenstra, 2021). Thus, as these youth develop into frank adolescents, the influence of FPE may also become more important.

A still possible third (and not mutually exclusive) possible explanation for FPE not accounting for unique variance in social anxiety beyond FNE within the present sample is that the FPES-Child (and BFNE-Child) items warrant some fine-tuning and additional psychometric investigation. This latter explanation is consistent with the relatively lower, albeit adequate internal consistency of the FPES-Child items in the present sample, as well as with the significant cross-loadings of two of the FPES-C items and one of the BFNE-C items on the opposite valence of fear of evaluation. To this end, we hope that the present study serves as a springboard for additional work examining FPE and the bivalent fear of evaluation model of social anxiety in children.

Additionally, it was unexpected that child -rated FPE did not relate significantly to parent-rated social anxiety. However, abundant research has shown that multiple informants achieve lower levels of correspondence for reports about behaviors that are more subtle/less salient to observe (e.g., anxiety) relative to behaviors that are easier to observe (e.g., conduct problems; e.g., see Achenbach, McConaughy, & Howell, 1987; De Los Reyes et al., 2015), and this has borne out in prior research on FPE and the bivalent fear of evaluation model in adolescents (e.g., see Karp et al., 2018, pp. 2156). Similarly, it may be that behaviors reflecting FNE may be easier for parents and children to observe than behaviors reflecting FPE, thus leading to lower correspondence between child rated FPE and parent-rated and child-rated reports of social anxiety relative to their reports of FNE. We were unable to test this possibility in the present study, and thus, recommend that future work directly examines this possibility and potential mechanisms therein.

Our findings have important theoretical and clinical implications. First, the ability to examine the developmental implications of the bivalent fear of evaluation model of social anxiety requires that measures of FPE and FNE shows internal structural validity, as well as convergent and divergent validity with critical clinical outcomes. Our findings indicate that FPE and FNE are both important constructs to social anxiety in childhood, and thus extend support for the bivalent fear of evaluation model from a developmental perspective. Second, future research should examine FPE and FNE in pediatric samples of social anxiety disorder (SAD) patients – for example, this could examine whether FPES-C and BFNE-C scores: adequately predict SAD diagnosis and demonstrate sensitivity to SAD treatment response, as has been observed in treatment outcome research with adults (e.g., see Weeks et al., 2012).

Several limitations to the present study exist. First, our participants were drawn from the first wave of a prospective three-year longitudinal study examining associations between development and reward function – our participants were not seeking treatment for SAD/did not undergo diagnostic assessment, and we were thus unable to examine predictive validity of the FPES-C or BFNE-C scores for SAD diagnostic status. Second, although this is the first study to examine a self-report version of the FPES for children, our design was cross-sectional – we strongly encourage researchers to add the FPES-C and BFNE-C to longitudinal designs with child samples. In particular, as already noted above, future research should examine FPES-C and BFNE-C responses in treatment monitoring and treatment outcomes in clinical work with child-aged SAD patients. Third, it is necessary to extend the validation of the FPES-C and BFNE-C through the inclusion of multimethod data sources. This may also benefit from examining the utility of the FPES-C and BFNE-C relative to alternative measures of these constructs (using different measures of FPE and FNE) to examine whether different items or conceptualization of item content could influence associations with clinical outcomes.

In conclusion, overall, our findings suggest that the FPES-C and BFNE-C can reliably and validly assess socio-evaluative fears among children, and that these fears are associated with social anxiety. These findings have important implications for extending research and theory on socio-evaluative fears and social anxiety in pediatric samples. However, it bears noting that our initial FPES-C as reported here may benefit from improvements – and we strongly encourage future researchers to take up this torch and carry it forward. Furthermore, our findings provide additional support for the bivalent fear of evaluation model of social anxiety, and begin to shed some initial light on developmental implications therein – nevertheless, the route forward from here is long (and hopefully fruitful), and we look forward to future research that navigates from here on the basis of the excellent findings from Poole and colleagues (2022) as well as our own.

Declaration of Competing Interest

None.

Data Availability

Data will be made available on request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.janxdis.2023.102784.

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